

Equilibrium of Rigid Bodies and Beam Analysis

High-Level Overview

In Engineering Mechanics, **equilibrium** is the state of a rigid body in which all surrounding forces and moments are perfectly balanced. When a body is in equilibrium, it experiences zero acceleration, meaning it either remains completely at rest (static equilibrium) or moves at a constant velocity (dynamic equilibrium).

To maintain equilibrium, a structural member (such as a beam) must be supported by physical constraints—known as **supports**—that exert **support reactions** to counteract external **loads**. Analyzing these loads, replacing distributed forces with equivalent point loads, and setting up static equilibrium equations allows engineers to determine the internal and external forces maintaining structural integrity.

Conditions of Static Equilibrium

For a coplanar (2D) system of forces acting on a rigid body, the resultant force R and net moment M must equal zero. This yields three fundamental scalar equations of static equilibrium:

1. Horizontal Force Balance:

$$\Sigma F_x = 0$$

(The sum of all rightward forces equals the sum of all leftward forces.)

2. Vertical Force Balance:

$$\Sigma F_y = 0$$

(The sum of all upward forces equals the sum of all downward forces.)

3. Moment Balance:

$$\Sigma M = 0$$

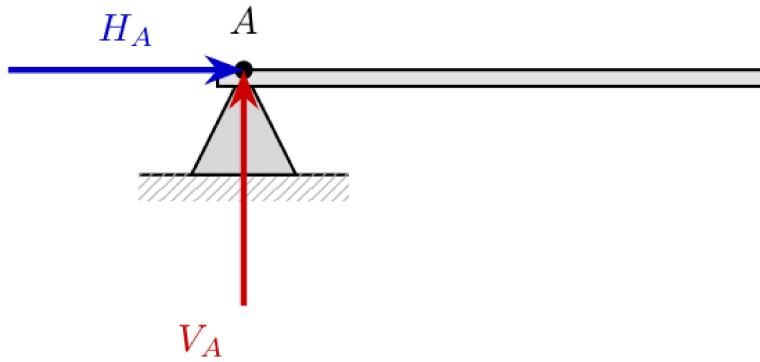
(The sum of all counter-clockwise moments about any point equals the sum of all clockwise moments about that same point.)

Types of Supports and Support Reactions

Supports prevent specific linear displacements and rotational movements. Every constrained motion generates a corresponding **reaction force or moment**.

1. Hinge (Pin) Support

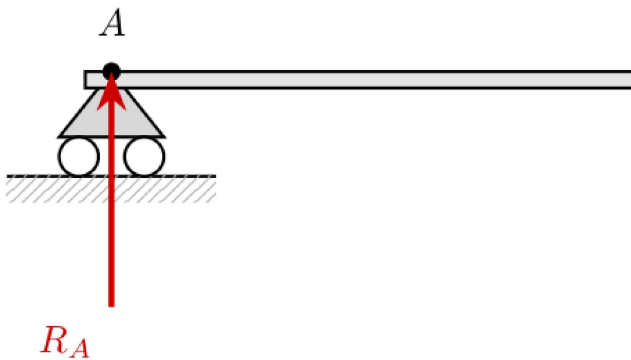
- **Motion Restricted:** Prevents horizontal and vertical translation. Allows rotation.
- **Reactions Generated:** Two linear reaction components:
 - Horizontal reaction force (H_A)
 - Vertical reaction force (V_A)



Visual Description: A horizontal beam resting on a pin/hinge support at end point A . The hinge support exerts two independent force components on the beam: a horizontal force vector H_A pointing rightward and a vertical force vector V_A pointing upward.

2. Roller Support

- **Motion Restricted:** Prevents movement perpendicular to the supporting surface. Allows movement parallel to the surface and allows rotation.
- **Reactions Generated:** One single reaction force (R_A) acting normal (perpendicular) to the supporting contact surface.



Visual Description: A horizontal beam supported at end A by a roller assembly resting on a flat surface. The roller exerts a single vertical reaction force vector R_A directed perpendicular to the surface upwards into the beam.

Types of Applied Structural Loads

Distributed loads act along a continuous length of a structural member rather than at a single discrete point. For calculation purposes, every distributed load is replaced by an **equivalent point load** equal to its area, applied at its **centroid** (center of gravity).

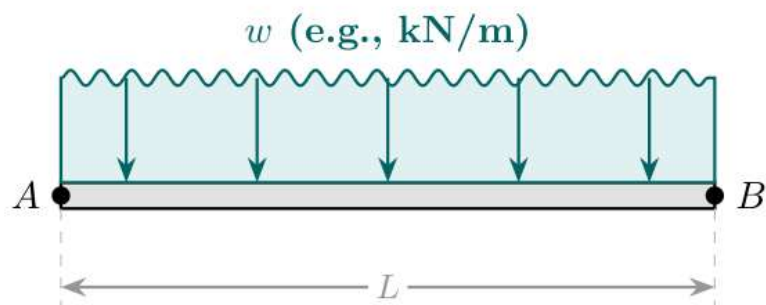
1. Uniformly Distributed Load (UDL)

A UDL spreads an equal amount of load per unit length across a span.

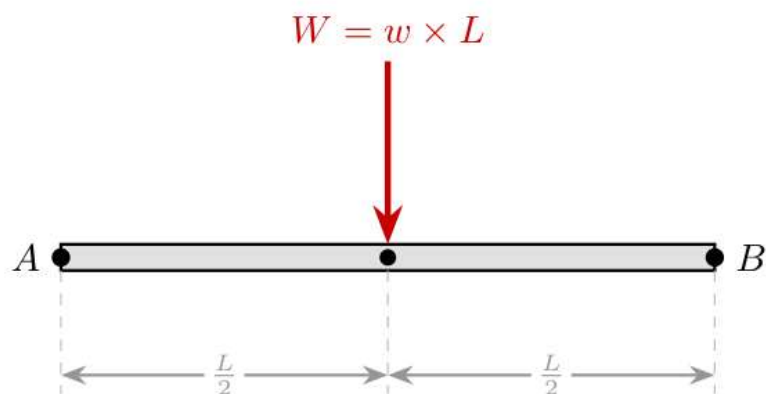
- **Intensity:** w (e.g., kN/m)
- **Span Length:** L
- **Equivalent Point Load (W):**

$$W = w \times L$$

- **Point of Application:** At the midpoint of the span ($\frac{L}{2}$ from either end).



⇓ Equivalent Conversion



Visual Description: Top diagram shows a continuous wave pattern across span length L representing a UDL of intensity w . Bottom diagram shows the equivalent point load of magnitude $W = w \times L$ acting vertically downward at a distance $\frac{L}{2}$ from either end.

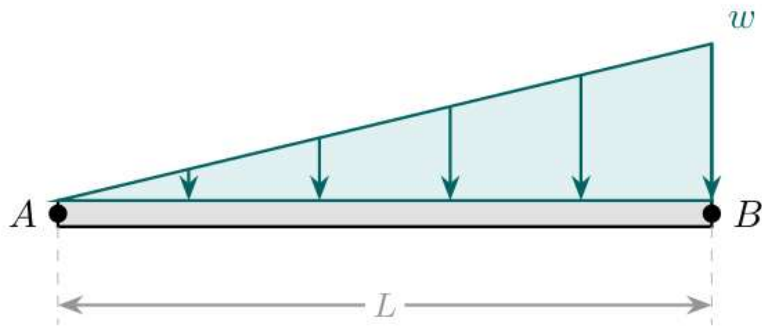
2. Uniformly Varying Load (UVL)

A UVL varies linearly from zero intensity at one end to a peak intensity w at the other (triangular shape).

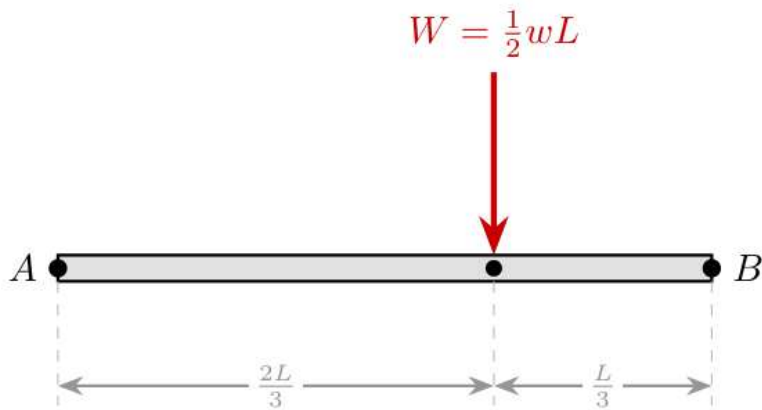
- **Peak Intensity:** w (e.g., kN/m)
- **Span Length:** L
- **Equivalent Point Load (W):** Equal to the area of the triangular distribution:

$$W = \frac{1}{2} \times w \times L$$

- **Point of Application:** Located at the centroid of the triangle:
 - $\frac{L}{3}$ from the larger (peak) side.
 - $\frac{2L}{3}$ from the vertex (zero load) side.



⇓ Equivalent Conversion



Visual Description: Top diagram displays a right-triangular load distribution along span L , starting at zero at point A and reaching maximum intensity w at point B . Bottom diagram shows the equivalent concentrated load $W = \frac{1}{2}wL$ acting downwards at a distance $\frac{L}{3}$ from the maximum load edge B .

3. Trapezoidal Load

A trapezoidal load varies linearly between two non-zero intensity values (w_1 and w_2). It is solved by breaking it into two simpler shapes: a rectangular UDL and a triangular UVL.

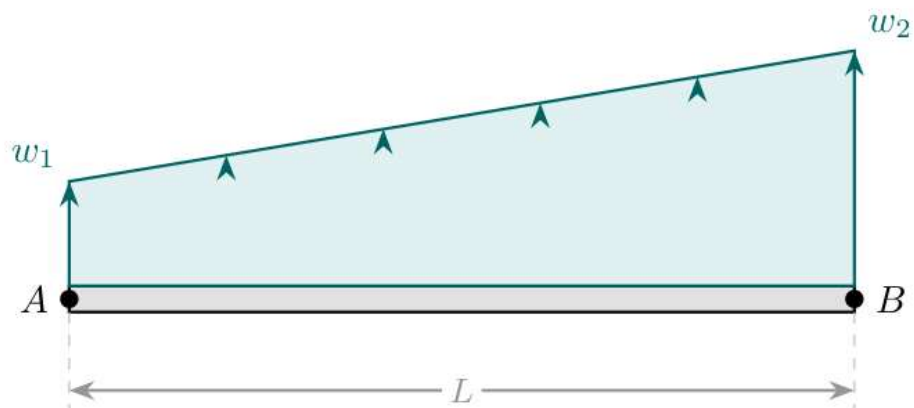
- **Components:**

- i. **Rectangular Part (UDL):** Intensity w_1 over length L .

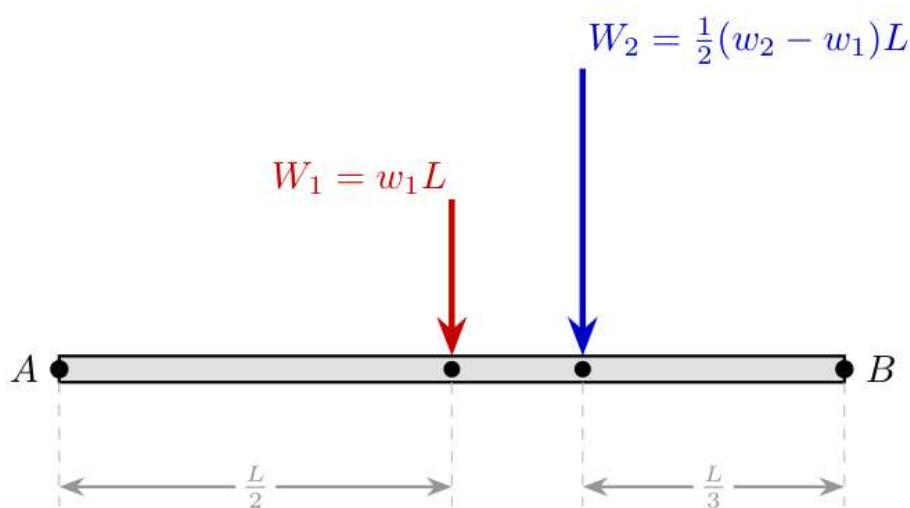
- Equivalent Load: $W_1 = w_1 \times L$, applied at distance $\frac{L}{2}$ from end A .

- ii. **Triangular Part (UVL):** Intensity $(w_2 - w_1)$ over length L .

- Equivalent Load: $W_2 = \frac{1}{2} \times (w_2 - w_1) \times L$, applied at distance $\frac{L}{3}$ from end B .



⇓ Equivalent Conversion



Visual Description: Top diagram displays a trapezoidal load starting at intensity w_1 at end A and rising to w_2 at end B over span L . Bottom diagram shows its resolution into two equivalent concentrated forces: rectangular force $W_1 = w_1 L$ acting at $\frac{L}{2}$ and triangular force $W_2 = \frac{1}{2}(w_2 - w_1)L$ acting at distance $\frac{L}{3}$ from B .

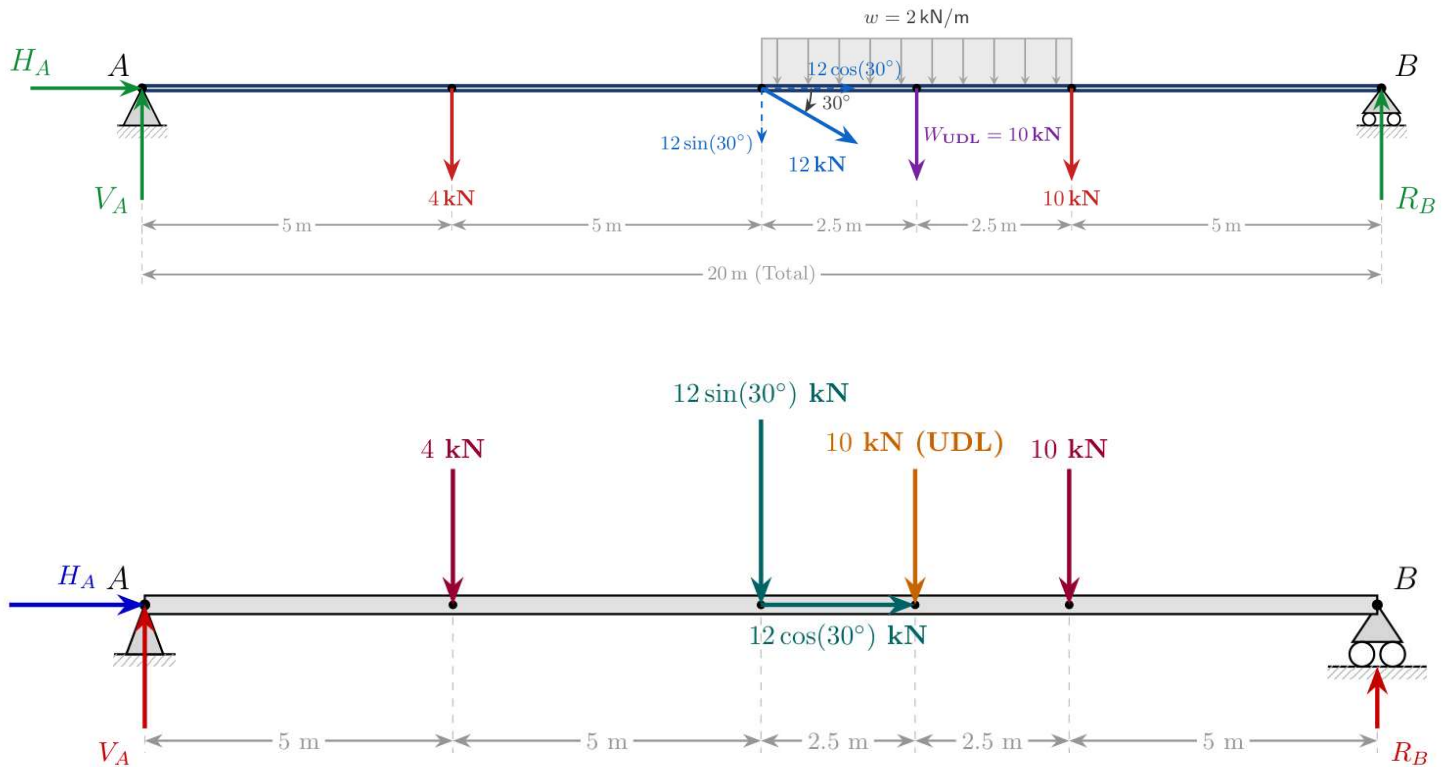
Worked Example: Calculating Beam Support Reactions

Problem Statement

A horizontal beam AB of total length 20 m is hinged at end A and supported on rollers at end B . The beam is subjected to the following loading configuration:

1. A vertical point load of 4 kN downward at 5 m from A .
2. An inclined force of 12 kN pointing downward and rightward at an angle of 30° to the horizontal beam at 10 m from A .
3. A Uniformly Distributed Load (UDL) of intensity 2 kN/m spanning a length of 5 m (between 10 m and 15 m from A).
4. A vertical point load of 10 kN downward at 15 m from A .

Find the support reactions at ends A and B .



Visual Description: Free Body Diagram of beam AB (20 m long). Hinge reactions H_A (rightward) and V_A (upward) act at A . Roller reaction R_B (upward) acts at B . Moving rightward: a 4 kN downward force at 5 m; an inclined force resolved into $12 \cos(30^\circ)$ rightward and $12 \sin(30^\circ)$ downward at 10 m; an equivalent 10 kN vertical downward force (from UDL) acting at 12.5 m; a 10 kN point load at 15 m.

Detailed Step-by-Step Solution

Step 1: Resolve the Inclined Force

The 12 kN force at an angle of 30° to the horizontal is resolved into components:

- Horizontal component (rightward):

$$12 \cos(30^\circ) = 12 \times 0.8660 = 10.392 \text{ kN} \quad (\rightarrow)$$

- Vertical component (downward):

$$12 \sin(30^\circ) = 12 \times 0.5 = 6 \text{ kN} \quad (\downarrow)$$

Step 2: Convert the UDL into an Equivalent Point Load

- Intensity $w = 2 \text{ kN/m}$, Span $L_{\text{UDL}} = 5 \text{ m}$
- Magnitude of equivalent point load:

$$W_{\text{UDL}} = 2 \text{ kN/m} \times 5 \text{ m} = 10 \text{ kN} \quad (\downarrow)$$

- Location: Midpoint of the UDL span:

$$\text{Distance from } A = 10 \text{ m} + \frac{5 \text{ m}}{2} = 12.5 \text{ m}$$

Step 3: Apply the Equilibrium Equation $\Sigma F_x = 0$

Summing all horizontal forces acting on the beam:

$$H_A + 12 \cos(30^\circ) = 0$$

$$H_A + 10.392 = 0 \implies H_A = -10.392 \text{ kN}$$

(The negative sign indicates that horizontal reaction force H_A acts leftward $\leftarrow$, opposite to the initially assumed direction.)

$$\mathbf{H_A = 10.392 \text{ kN} } \quad (\leftarrow)$$

Step 4: Apply Moment Equilibrium $\Sigma M_A = 0$

Taking moments about hinge point A , setting counter-clockwise (CCW) moments as positive (+) and clockwise (CW) moments as negative (-):

1. 4 kN Point Load:

$$\text{Moment} = -(4 \times 5) = -20 \text{ kN} \cdot \text{m}$$

2. Vertical Component of 12 kN Force ($12 \sin 30^\circ = 6 \text{ kN}$):

$$\text{Moment} = -(12 \sin 30^\circ \times 10) = -(6 \times 10) = -60 \text{ kN} \cdot \text{m}$$

3. Horizontal Component of 12 kN Force ($12 \cos 30^\circ$):

$$\text{Passes through point } A \implies \text{Moment} = 0$$

4. Equivalent Point Load of UDL (10 kN):

$$\text{Moment} = -(10 \times 12.5) = -125 \text{ kN} \cdot \text{m}$$

5. 10 kN Point Load:

$$\text{Moment} = -(10 \times 15) = -150 \text{ kN} \cdot \text{m}$$

6. Roller Support Reaction R_B (Upward at 20 m):

$$\text{Moment} = +(R_B \times 20)$$

Summing moments about A :

$$\Sigma M_A = -(4 \times 5) - (12 \sin 30^\circ \times 10) - (10 \times 12.5) - (10 \times 15) + (R_B \times 20) = 0$$

$$-20 - 60 - 125 - 150 + 20R_B = 0$$

$$-355 + 20R_B = 0$$

$$20R_B = 355$$

$$\mathbf{R_B = 17.75 \text{ kN} } \quad (\uparrow)$$

Step 5: Apply the Equilibrium Equation $\Sigma F_y = 0$

Summing all vertical forces acting on the beam:

$$V_A - 4 - 12 \sin(30^\circ) - 10_{\text{UDL}} - 10 + R_B = 0$$

$$V_A - 4 - 6 - 10 - 10 + 17.75 = 0$$

$$V_A - 30 + 17.75 = 0$$

$V_A - 12.25 = 0$

$V_A = 12.25 \text{ kN} \quad (\uparrow)$

Load & Support Reaction Summary Table

Support / Load Type	Primary Motion / Distribution Characteristics	Equivalent Point Load	Centroidal Location	Reaction Forces
Hinge Support	Restrains x and y linear translations	N/A	N/A	H_A (Horizontal), V_A (Vertical)
Roller Support	Restrains normal linear translation	N/A	N/A	R_A (Normal to contact surface)
UDL	Uniform intensity w across span L	$W = w \cdot L$	$\frac{L}{2}$ from end	N/A
UVL	Linear intensity variation $0 \rightarrow w$ across L	$W = \frac{1}{2}wL$	$\frac{L}{3}$ from peak side	N/A
Trapezoidal Load	Linear intensity variation $w_1 \rightarrow w_2$ across L	$W_1 = w_1L, W_2 = \frac{1}{2}(w_2 - w_1)L$	$\frac{L}{2}$ for UDL part, $\frac{L}{3}$ for UVL part	N/A